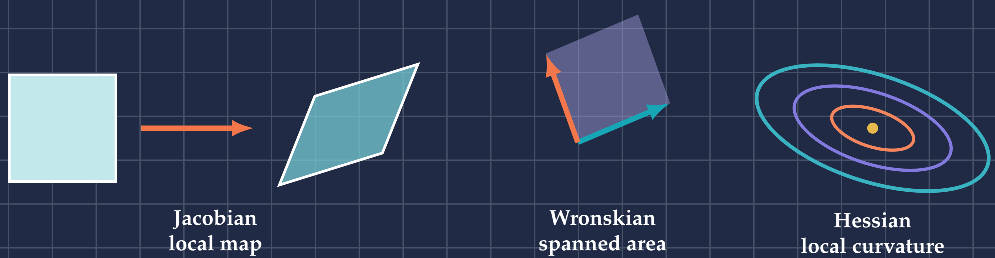
A VISUAL TEXTBOOK OF DERIVATIVE ARRAYS

Jacobians, Wronskians & Hessians

Linear maps, independence, and curvature



Theory, calculations, geometry, and worked examples

DF $\det \mathcal{W}$ $D(\nabla f)$

Three derivative objects, one geometric language

Each object in this book is an array made from derivatives. The entries matter, but the real payoff comes from reading what the whole array does.

Jacobian DF	**Wronskian** $\mathcal{W}$	**Hessian** $D(\nabla f)$
first derivatives of a vector map	functions and successive derivatives	second derivatives of a scalar field

three different assemblies of derivative information

object	calculation	geometric question
Jacobian	arrange all first partial derivatives	How does a small vector, area, or volume transform?
Wronskian	take a determinant of derivative columns	Do evolving functions provide independent directions?
Hessian	differentiate the gradient	How does slope change, and which way does a surface curve?

The recurring workflow

1. identify inputs and outputs → 2. differentiate entry by entry → 3. assemble with a fixed convention → 4. interpret the whole array

Notation used throughout

Vectors are columns. For $\boldsymbol{F} : \mathbb{R}^n \to \mathbb{R}^m$, the Jacobian is the $m \times n$ matrix whose (i, j) entry is $\partial F_i / \partial x_j$. Gradients are columns, so the Hessian is $D(\nabla f)$. Other books may transpose one of these conventions; the underlying linear map is the same once dimensions are handled consistently.

A naming trap to avoid

"The Jacobian" may mean the matrix $D\boldsymbol{F}$ or, in change-of-variables contexts, its determinant $\det D\boldsymbol{F}$. This book always says *Jacobian matrix* or *Jacobian determinant* when ambiguity matters.

The Jacobian: the derivative of a vector map

In one variable, a derivative is the best local multiplier:

$$f(x + h) \approx f(x) + f'(x)h.$$

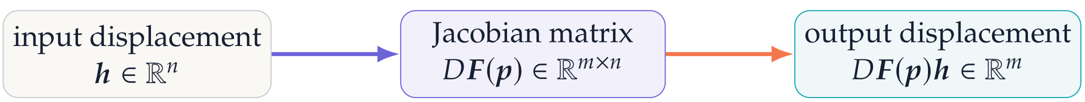
For a map $\boldsymbol{F} : \mathbb{R}^n \longrightarrow \mathbb{R}^m$, a single multiplier is no longer enough. The best local linear map is a matrix.

Definition

If $\boldsymbol{F} = (F_1, \ldots, F_m)$ and $\boldsymbol{x} = (x_1, \ldots, x_n)$, then

$$D\boldsymbol{F}(\boldsymbol{x}) = \mathbf{J}_{\boldsymbol{F}}(\boldsymbol{x}) = \begin{bmatrix} \dfrac{\partial F_1}{\partial x_1} & \cdots & \dfrac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial F_m}{\partial x_1} & \cdots & \dfrac{\partial F_m}{\partial x_n} \end{bmatrix}.$$

Rows correspond to output components; columns correspond to input variables.

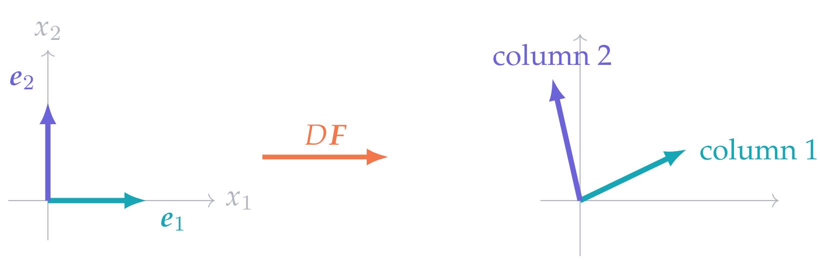


Why the columns matter

Let $\boldsymbol{e}_j$ be the jth coordinate direction. Then

$$D\boldsymbol{F}(\boldsymbol{p})\boldsymbol{e}_j = \frac{\partial \boldsymbol{F}}{\partial x_j}(\boldsymbol{p}),$$

so column j tells where the jth input axis points after the local map.



Dimension check

An $m \times n$ Jacobian multiplies an $n \times 1$ input displacement and produces an $m \times 1$ output displacement. This quick check catches many transposition errors before any arithmetic begins.

Calculating a Jacobian, step by step

Consider

$$\boldsymbol{F}(x, y) = \begin{bmatrix} x^2 - y \\ xy + y^2 \end{bmatrix}.$$

Four reliable steps

1. List outputs: $F_1 = x^2 - y$ and $F_2 = xy + y^2$.
2. List inputs in order: x, y.
3. Differentiate each output with respect to each input.
4. Put output i in row i and input j in column j.

$$\frac{\partial F_1}{\partial x} = 2x, \quad \frac{\partial F_1}{\partial y} = -1, \qquad \frac{\partial F_2}{\partial x} = y, \quad \frac{\partial F_2}{\partial y} = x + 2y.$$

Therefore

Jacobian matrix

$$D\boldsymbol{F}(x, y) = \begin{bmatrix} 2x & -1 \\ y & x + 2y \end{bmatrix}.$$

At $\boldsymbol{p} = (1, 2)$,

$$D\boldsymbol{F}(1, 2) = \begin{bmatrix} 2 & -1 \\ 2 & 5 \end{bmatrix}, \qquad \det D\boldsymbol{F}(1, 2) = 12.$$

$$\begin{array}{cc} & \begin{array}{cc} x & y \end{array} \\ \text{outputs} & \begin{bmatrix} 2x & -1 \\ y & x + 2y \end{bmatrix} \end{array}$$

row 1: derivatives of F_1
row 2: derivatives of F_2

A numerical local prediction

Here $\boldsymbol{F}(1, 2) = (-1, 6)$. For $\boldsymbol{h} = (0.02, -0.01)$,

$$D\boldsymbol{F}(1, 2)\boldsymbol{h} = \begin{bmatrix} 2 & -1 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} .02 \\ -.01 \end{bmatrix} = \begin{bmatrix} .05 \\ -.01 \end{bmatrix}.$$

Thus $\boldsymbol{F}(1.02, 1.99)$ should be close to $(-.95, 5.99)$. Direct evaluation gives $(-.9496, 5.9899)$; the small discrepancy is the neglected nonlinear part.

Sanity checks

The matrix has two rows and two columns because the map has two outputs and two inputs. Each entry is a scalar. Evaluation happens only after differentiation.

Local linearization: what the Jacobian actually does

Differentiability means that near $\boldsymbol{p}$,

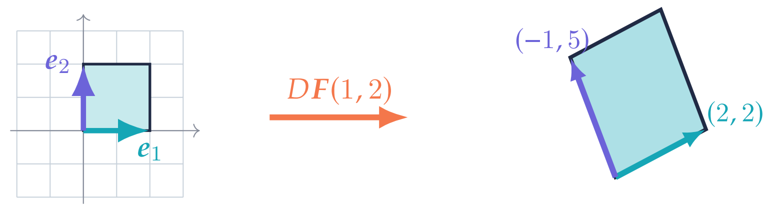
First-order model

$$\boxed{\boldsymbol{F}(\boldsymbol{p}+\boldsymbol{h}) = \boldsymbol{F}(\boldsymbol{p}) + D\boldsymbol{F}(\boldsymbol{p})\boldsymbol{h} + o(\|\boldsymbol{h}\|)}.$$

The translation $\boldsymbol{F}(\boldsymbol{p})$ locates the image point. The matrix $D\boldsymbol{F}(\boldsymbol{p})$ controls local shape, scale, shear, and orientation.

For the previous example at $(1,2)$, the two coordinate directions become

$$\boldsymbol{e}_1 \longmapsto \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \boldsymbol{e}_2 \longmapsto \begin{bmatrix} -1 \\ 5 \end{bmatrix}.$$



tiny input square **local output parallelogram**

Derivative error shrinks faster than the step

The symbol $o(\|\boldsymbol{h}\|)$ means

$$\frac{\|\boldsymbol{F}(\boldsymbol{p}+\boldsymbol{h}) - \boldsymbol{F}(\boldsymbol{p}) - D\boldsymbol{F}(\boldsymbol{p})\boldsymbol{h}\|}{\|\boldsymbol{h}\|} \longrightarrow 0 \quad \text{as } \boldsymbol{h} \to \boldsymbol{0}.$$

That is stronger than merely saying the error is small: relative to the step, the linearization becomes arbitrarily accurate.

The geometric reading

Zoom in on a differentiable map. Curved grid lines become nearly straight, and a small coordinate box becomes the parallelogram spanned by the Jacobian columns. The Jacobian is what remains after the nonlinear map is viewed under a microscope.

The Jacobian determinant: local area and orientation

For a square Jacobian $DF : \mathbb{R}^n \to \mathbb{R}^n$, the determinant measures signed local volume scaling.

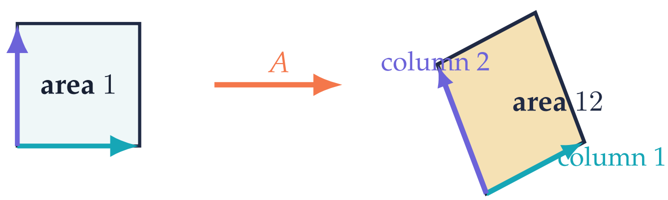
Area and volume law

$$\text{small output volume} \approx |\det DF(p)| \times \text{small input volume}.$$

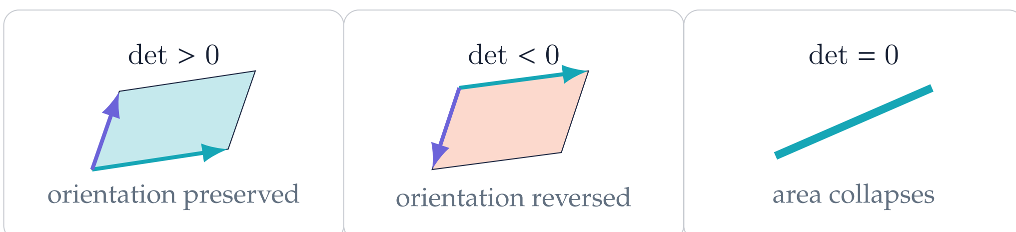
The sign records orientation. A negative determinant reverses handedness; a zero determinant collapses at least one local direction.

For $A = \begin{bmatrix} 2 & -1 \\ 2 & 5 \end{bmatrix}$, the column vectors span an area

$$\det A = 2(5) - (-1)(2) = 12.$$



Three determinant regimes



Determinant zero is a local warning

If $\det DF(p) = 0$, the inverse function theorem cannot guarantee a smooth local inverse at p. A zero determinant does not by itself describe the entire global map; it identifies a local loss of full-dimensional information.

Change of variables: why polar area contains r

The polar-coordinate map is

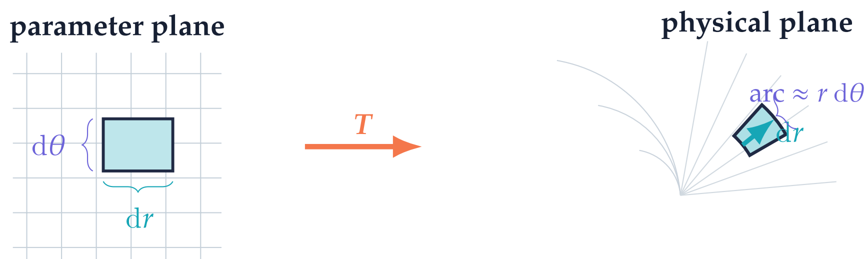
$$\boldsymbol{T}(r,\theta) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix}.$$

Its Jacobian matrix and determinant are

Polar Jacobian

$$D\boldsymbol{T}(r,\theta) = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}, \qquad \det D\boldsymbol{T}(r,\theta) = r.$$

The first column is a unit radial direction. The second is tangent to the circle and has length r. A tiny rectangle of side lengths $\mathrm{d}r$ and $\mathrm{d}\theta$ therefore becomes a curved sector with area approximately $r\,\mathrm{d}r\,\mathrm{d}\theta$.



The integration rule

If $\boldsymbol{x} = \boldsymbol{T}(\boldsymbol{u})$ is one-to-one on the region of interest, then

$$\int_{T(U)} g(\boldsymbol{x})\,\mathrm{d}\boldsymbol{x} = \int_U g(\boldsymbol{T}(\boldsymbol{u}))\,|\det D\boldsymbol{T}(\boldsymbol{u})|\,\mathrm{d}\boldsymbol{u}.$$

For polar coordinates this becomes

$$\iint g(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint g(r\cos\theta, r\sin\theta)\,r\,\mathrm{d}r\,\mathrm{d}\theta.$$

Why the absolute value?

Integration measures unsigned area or volume. The determinant sign records orientation, but physical volume uses $|\det D\boldsymbol{T}|$.

Chain rule and inverse maps

Suppose $\boldsymbol{F} : \mathbb{R}^n \to \mathbb{R}^m$ and $\boldsymbol{G} : \mathbb{R}^m \to \mathbb{R}^k$. Local linear maps compose in the same order as the functions:

Multivariable chain rule

$$D(\boldsymbol{G} \circ \boldsymbol{F})(\boldsymbol{x}) = D\boldsymbol{G}(\boldsymbol{F}(\boldsymbol{x}))\, D\boldsymbol{F}(\boldsymbol{x}).$$

The rightmost Jacobian acts first. Matrix dimensions are $(k \times m)(m \times n) = k \times n$.

$$\begin{tikzcd} \begin{matrix}\mathbb{R}^n \\ \boldsymbol{h}\end{matrix} \arrow[r, "D\boldsymbol{F}"] \arrow[rr, bend right, "D(\boldsymbol{G} \circ \boldsymbol{F})"'] & \begin{matrix}\mathbb{R}^m \\ D\boldsymbol{F}\,\boldsymbol{h}\end{matrix} \arrow[r, "D\boldsymbol{G}"] & \begin{matrix}\mathbb{R}^k \\ D\boldsymbol{G}\,D\boldsymbol{F}\,\boldsymbol{h}\end{matrix} \end{tikzcd}$$

Scalar outer function: gradients emerge

If $g : \mathbb{R}^m \to \mathbb{R}$, then

$$\nabla(g \circ \boldsymbol{F})(\boldsymbol{x}) = D\boldsymbol{F}(\boldsymbol{x})^{\mathsf{T}} \nabla g(\boldsymbol{F}(\boldsymbol{x})).$$

The transpose appears because gradients are represented as column vectors.

Derivative of a local inverse

If $\boldsymbol{F}$ has a differentiable local inverse and $D\boldsymbol{F}(\boldsymbol{p})$ is invertible, differentiate $\boldsymbol{F}^{-1}(\boldsymbol{F}(\boldsymbol{x})) = \boldsymbol{x}$:

$$D\boldsymbol{F}^{-1}(\boldsymbol{F}(\boldsymbol{p}))\, D\boldsymbol{F}(\boldsymbol{p}) = I.$$

Therefore

Inverse derivative

$$\boxed{D\boldsymbol{F}^{-1}(\boldsymbol{F}(\boldsymbol{p})) = [D\boldsymbol{F}(\boldsymbol{p})]^{-1}.}$$

A quick inverse calculation

If $D\boldsymbol{F}(\boldsymbol{p}) = \begin{bmatrix} 2 & -1 \\ 2 & 5 \end{bmatrix}$, then

$$D\boldsymbol{F}^{-1}(\boldsymbol{F}(\boldsymbol{p})) = \frac{1}{12} \begin{bmatrix} 5 & 1 \\ -2 & 2 \end{bmatrix}.$$

Multiplying the two matrices gives the identity, which is the fastest check.

Part I in one sentence

The Jacobian is the best local linear map; its columns move coordinate directions, its determinant scales oriented volume, and matrix multiplication expresses the chain rule.

The Wronskian: independence encoded by derivatives

Given n functions $y_1, \dots, y_n$ with enough derivatives, arrange each function and its first $n-1$ derivatives in a column.

Definition

$$\mathcal{W}(y_1, \dots, y_n)(t) = \det \begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}.$$

If the functions satisfy a constant relation $c_1 y_1 + \cdots + c_n y_n = 0$, then differentiating it $n-1$ times gives a homogeneous matrix equation. A nonzero determinant rules out any nonzero coefficient vector.

$$\text{successive derivatives}\left\{\underbrace{\begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix}}_{\text{one function per column}}\right. \qquad \boxed{\begin{array}{c} \det \neq 0 \\ \Downarrow \\ \text{columns independent} \end{array}}$$

How to calculate it

For two functions,

$$\mathcal{W}(y_1, y_2) = \det \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} = y_1 y_2' - y_2 y_1'.$$

For $y_1 = e^t$ and $y_2 = te^t$,

$$\begin{aligned} \mathcal{W}(e^t, te^t) &= \det \begin{bmatrix} e^t & te^t \\ e^t & (1+t)e^t \end{bmatrix} \\ &= (1+t)e^{2t} - te^{2t} = e^{2t} \neq 0. \end{aligned}$$

Reliable calculation order

1. Put the original functions in the first row.
2. Differentiate the whole previous row to make the next row.
3. Stop after derivative order $n-1$.
4. Compute and simplify the determinant; then interpret where it vanishes.

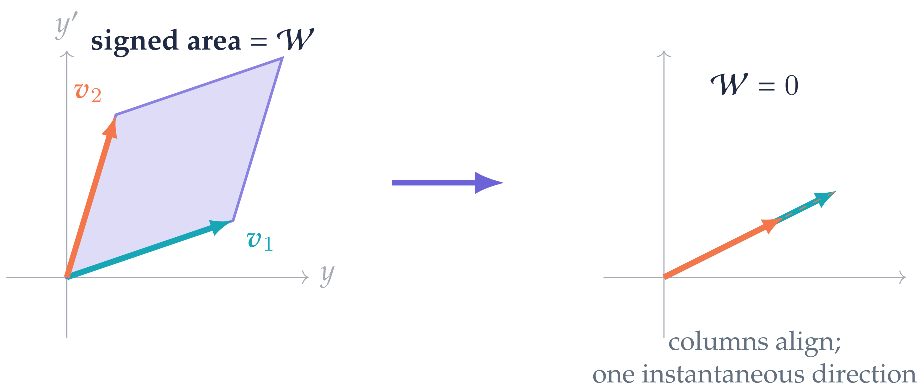
What nonzero proves. If $\mathcal{W}(t_0) \neq 0$ at one point, the functions are linearly independent on their common interval. The converse requires the hypotheses stated on page 12.

The geometry of a two-function Wronskian

For two functions define their state columns

$$v_1(t) = \begin{bmatrix} y_1(t) \\ y_1'(t) \end{bmatrix}, \qquad v_2(t) = \begin{bmatrix} y_2(t) \\ y_2'(t) \end{bmatrix}.$$

Then $\mathcal{W}(y_1, y_2) = \det[v_1\ v_2]$: it is the signed area of the parallelogram spanned by the two states.

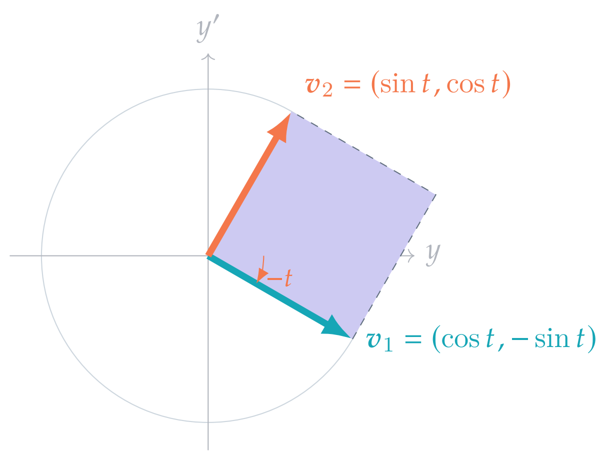


A perfect rotating example

Take $y_1 = \cos t$ and $y_2 = \sin t$:

$$\mathcal{W}(\cos t, \sin t) = \det \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix} = 1.$$

The two state columns are always perpendicular unit vectors. They rotate, but the area between them remains exactly one.



Geometric meaning

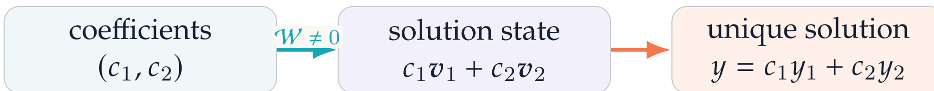
The Wronskian does not merely compare function values. It compares enough derivative information to build states. Nonzero state volume means the functions provide distinct directions in the space of initial data.

Why the Wronskian is central in linear ODEs

For the second-order homogeneous equation

$$y'' + p(t)y' + q(t)y = 0,$$

two solutions form a fundamental pair exactly when their state columns span the two-dimensional state space.



Abel's identity

Let $W = y_1y_2' - y_2y_1'$. Differentiating and using the ODE gives

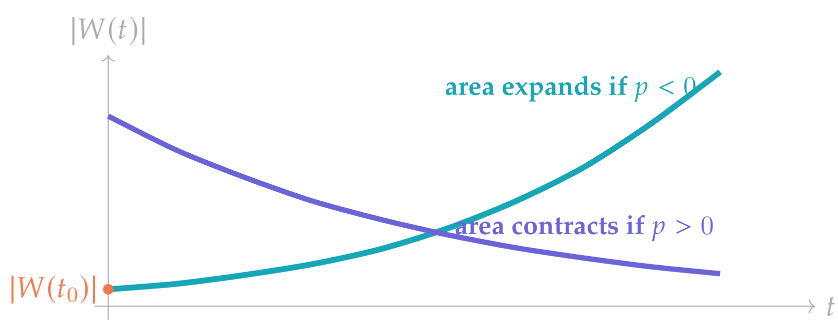
$$\begin{aligned} W' &= y_1y_2'' - y_2y_1'' \\ &= y_1(-py_2' - qy_2) - y_2(-py_1' - qy_1) \\ &= -p(y_1y_2' - y_2y_1') = -pW. \end{aligned}$$

Therefore

Abel's formula

$$\boxed{W(t) = W(t_0)\exp\left(-\int_{t_0}^{t} p(s)\,\mathrm{d}s\right).}$$

The exponential never vanishes. Hence a solution Wronskian is either zero everywhere or nonzero everywhere on the interval.



Example: $y'' - 3y' + 2y = 0$

Solutions $y_1 = e^t$ and $y_2 = e^{2t}$ give

$$W = \det\begin{bmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{bmatrix} = e^{3t}.$$

Here $p = -3$, so Abel predicts $W(t) = W(0)e^{3t} = e^{3t}$ exactly.

Higher-order Wronskians and the essential caveat

For $y_1 = 1$, $y_2 = t$, and $y_3 = t^2$,

A triangular calculation

$$\mathcal{W}(1, t, t^2) = \det \begin{bmatrix} 1 & t & t^2 \\ 0 & 1 & 2t \\ 0 & 0 & 2 \end{bmatrix} = 2.$$

The triangular form makes the determinant the product of diagonal entries.

$$\underset{\text{state of } 1}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}} \longrightarrow \underset{\text{state of } t}{\begin{bmatrix} t \\ 1 \\ 0 \end{bmatrix}} \longrightarrow \underset{\text{state of } t^2}{\begin{bmatrix} t^2 \\ 2t \\ 2 \end{bmatrix}}$$

three independent directions in initial-data space

What zero does and does not mean

The precise logical rule

- If $W(t_0) \neq 0$, the functions are linearly independent.
- If arbitrary differentiable functions have $W \equiv 0$, independence is *not* ruled out without extra hypotheses.
- If the functions solve the same linear homogeneous ODE with continuous coefficients, then $W \equiv 0$ is equivalent to dependence.

An instructive counterexample on $(-1, 1)$ is

$$y_1(t) = t^2, \qquad y_2(t) = t|t|.$$

These differentiable functions are linearly independent: on positive and negative t, a constant relation imposes two different sign conditions. Yet direct calculation gives $\mathcal{W}(y_1, y_2) \equiv 0$.

Also, a Wronskian may vanish at an isolated point without implying dependence. For t^2 and t^3,

$$\mathcal{W}(t^2, t^3) = t^4,$$

which vanishes at $t = 0$ although the functions are independent.

Part II in one sentence

The Wronskian is the signed volume of derivative-state columns: nonzero volume certifies independence, and Abel's identity makes that test especially powerful for solutions of one linear homogeneous ODE.

The Hessian: the derivative of the gradient

For a scalar field $f : \mathbb{R}^n \to \mathbb{R}$, the gradient collects all first partial derivatives. Differentiating the gradient produces the Hessian.

Definition

$$\mathbf{H}_f(\boldsymbol{x}) = D(\nabla f)(\boldsymbol{x}) = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \cdots & \dfrac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \dfrac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

For $f \in C^2$, equality of mixed partials makes $\mathbf{H}_f$ symmetric.

$$\begin{array}{c} \text{scalar field} \\ f : \mathbb{R}^n \to \mathbb{R} \end{array} \xrightarrow{D} \begin{array}{c} \text{slope vector} \\ \nabla f : \mathbb{R}^n \to \mathbb{R}^n \end{array} \xrightarrow{D} \begin{array}{c} \text{curvature map} \\ D(\nabla f) = \mathbf{H}_f \end{array}$$

A complete calculation

Compute the gradient first and then take its Jacobian. This fixes the row and column order and gives two routes to the mixed partials.

Let

$$f(x, y) = x^2 y + 3xy^2 + \sin x.$$

First compute the gradient:

$$f_x = 2xy + 3y^2 + \cos x, \qquad f_y = x^2 + 6xy.$$

Now differentiate those components:

$$\begin{aligned} f_{xx} &= 2y - \sin x, & \qquad f_{xy} &= 2x + 6y, \\ f_{yx} &= 2x + 6y, & \qquad f_{yy} &= 6x. \end{aligned}$$

Thus

Hessian matrix

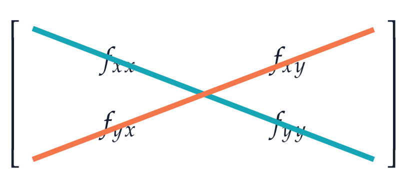
$$\mathbf{H}_f(x, y) = \begin{bmatrix} 2y - \sin x & 2x + 6y \\ 2x + 6y & 6x \end{bmatrix}.$$

At $(0, 1)$,

$$\mathbf{H}_f(0, 1) = \begin{bmatrix} 2 & 6 \\ 6 & 0 \end{bmatrix}.$$

off-diagonal

coupled change
between variables



diagonal

pure curvature
along coordinate axes

Second-order Taylor theory and directional curvature

The Hessian appears because the gradient itself changes from point to point. Near $\boldsymbol{p}$, a twice differentiable scalar field has the model

Second-order Taylor expansion

$$\boxed{f(\boldsymbol{p}+\boldsymbol{h}) \approx f(\boldsymbol{p}) + \nabla f(\boldsymbol{p})^{\top}\boldsymbol{h} + \frac{1}{2}\boldsymbol{h}^{\top}\mathbf{H}_f(\boldsymbol{p})\boldsymbol{h}.}$$

Constant + plane + quadratic bending is the local shape of the graph.

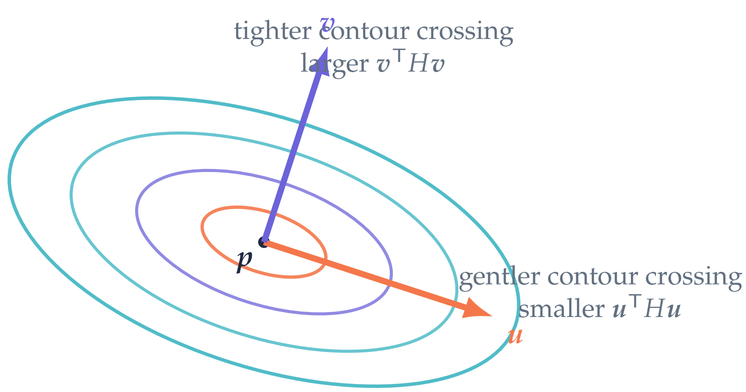
$f(\boldsymbol{p})$	$\nabla f(\boldsymbol{p})^{\top}\boldsymbol{h}$	$\frac{1}{2}\boldsymbol{h}^{\top}\mathbf{H}_f\boldsymbol{h}$
height constant term	tangent plane first-order tilt	curvature second-order bending

Curvature in a chosen direction

For a unit vector $\boldsymbol{u}$, restrict the surface to the line $\boldsymbol{p}+t\boldsymbol{u}$. Differentiating twice gives

Directional second derivative

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} f(\boldsymbol{p}+t\boldsymbol{u})\Big|_{t=0} = \boldsymbol{u}^{\top}\mathbf{H}_f(\boldsymbol{p})\boldsymbol{u}.$$



Because a symmetric Hessian has orthogonal eigenvectors, its eigenvectors give the principal curvature directions. Along a unit eigenvector $\boldsymbol{q}_i$,

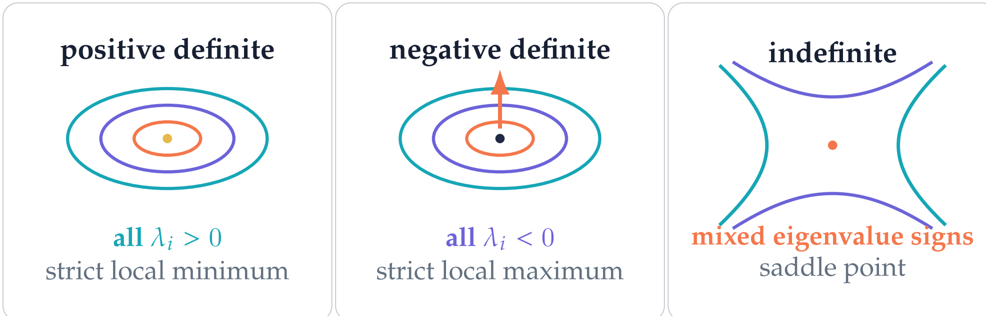
$$\boldsymbol{q}_i^{\top}\mathbf{H}_f\boldsymbol{q}_i = \lambda_i.$$

At a critical point

When $\nabla f(\boldsymbol{p}) = 0$, the tangent-plane term disappears. The quadratic form $\frac{1}{2}\boldsymbol{h}^{\top}H\boldsymbol{h}$ becomes the first informative local term, so the signs of the Hessian eigenvalues determine the basic shape.

Classifying critical points with the Hessian

At a critical point $\nabla f(\boldsymbol{p}) = 0$, inspect the eigenvalues of the symmetric Hessian.



The two-variable determinant test

For $H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$, define

$$\Delta = f_{xx} f_{yy} - f_{xy}^2 = \det H.$$

condition	Hessian type	conclusion at a critical point
$\Delta > 0$, $f_{xx} > 0$	positive definite	strict local minimum
$\Delta > 0$, $f_{xx} < 0$	negative definite	strict local maximum
$\Delta < 0$	indefinite	saddle point
$\Delta = 0$	semidefinite or degenerate	test is inconclusive

Rotated quadratic bowl

For

$$f(x, y) = x^2 + xy + 3y^2 - 4x + 2y,$$

the critical point solves $2x + y - 4 = 0$ and $x + 6y + 2 = 0$, giving $\boldsymbol{p} = (26/11, -8/11)$. The Hessian is

$$H = \begin{bmatrix} 2 & 1 \\ 1 & 6 \end{bmatrix}, \qquad \det H = 11 > 0, \quad f_{xx} = 2 > 0.$$

So $\boldsymbol{p}$ is a strict minimum. The eigenvalues $4 \pm \sqrt{5}$ are positive; their eigenvectors give the rotated axes of the elliptical contours.

When the test is silent

If a Hessian has a zero eigenvalue, higher-order terms may decide the shape. For example, both $x^4 + y^4$ and $x^4 - y^4$ have zero Hessian at the origin, yet the first has a minimum and the second a saddle.

Newton's method: using curvature to choose a step

To seek a critical point, linearize the gradient:

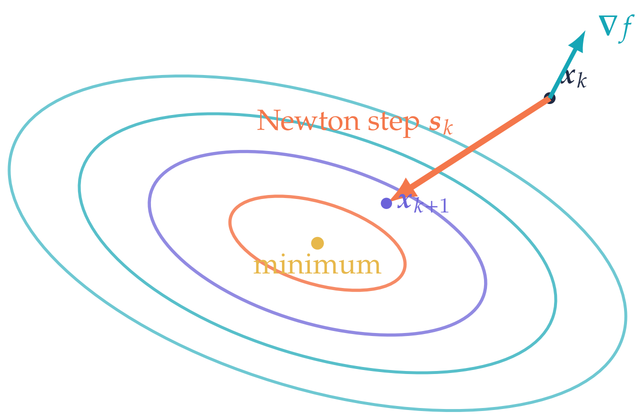
$$\nabla f(x+s) \approx \nabla f(x) + \mathbf{H}_f(x)s.$$

Set the model equal to zero and solve for the step.

Newton step

$$\boxed{\mathbf{H}_f(x_k)s_k = -\nabla f(x_k), \qquad x_{k+1} = x_k + s_k.}$$

In computation, solve the linear system; do not explicitly form H^{-1} unless there is a special reason.



Why quadratics are solved in one step

Let

$$f(x) = \frac{1}{2}x^\top Ax - b^\top x + c, \qquad A = A^\top.$$

Then $\nabla f = Ax - b$ and $H = A$. Newton gives

$$x_{k+1} = x_k - A^{-1}(Ax_k - b) = A^{-1}b,$$

the exact stationary point, regardless of the starting point.

Curvature can mislead

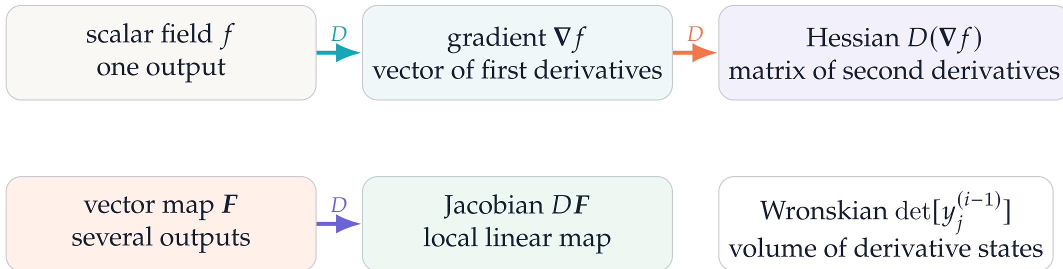
Far from a solution, the quadratic model may be poor. An indefinite or nearly singular Hessian can point uphill or create an enormous step. Practical methods use damping, line search, trust regions, or positive-definite Hessian approximations to control the move.

Beyond optimization

Hessians also appear in uncertainty approximations, sensitivity analysis, Laplace's method, mechanics, image processing, and stability theory. In each case, the same quadratic form measures how a first-order quantity changes locally.

The common structure: derivatives assembled for a purpose

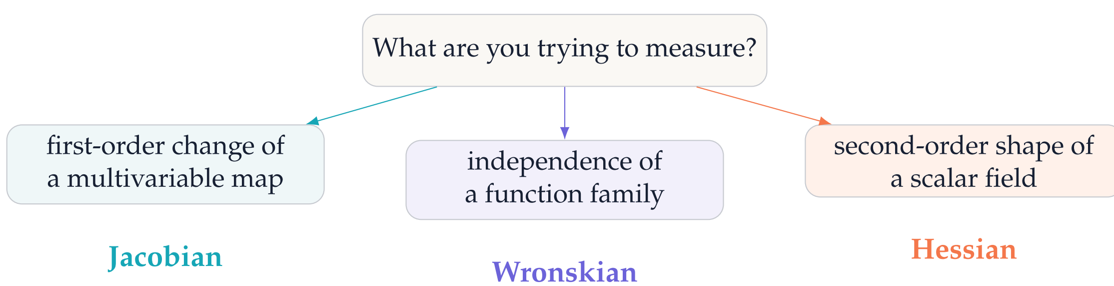
The three objects become easier to remember when their roles are placed on one calculus ladder.



What operation extracts the geometry?

array	operation	information revealed
$D\boldsymbol{F}$	multiply by a displacement	local direction, stretch, shear, and sensitivity
$D\boldsymbol{F}$	take a determinant when square	signed local area or volume factor
Wronskian matrix	take its determinant	independence of derivative states
$\mathbf{H}_f$	evaluate $\boldsymbol{u}^\top H\boldsymbol{u}$	curvature along direction $\boldsymbol{u}$
$\mathbf{H}_f$	compute eigenvalues/eigenvectors	principal curvatures and local shape

A decision guide



The unifying mental model

Differentiate first, then assemble derivatives so a familiar linear-algebra operation can answer the geometric question. Multiplication tracks local motion; determinants track spanned volume; quadratic forms and eigenvalues track curvature. Keep variable order fixed, check dimensions, and interpret a zero test cautiously.

Compact reference and practice

Jacobian
$[DF]_{ij} = \partial F_i/\partial x_j$
$\boldsymbol{F}(\boldsymbol{p}+\boldsymbol{h}) \approx \boldsymbol{F}(\boldsymbol{p}) + D\boldsymbol{F}(\boldsymbol{p})\boldsymbol{h}$
$|\det D\boldsymbol{F}|$ = local volume factor

Wronskian
column j = $(y_j, y_j', \ldots, y_j^{(n-1)})^\top$
$W \neq 0 \Rightarrow$ independence
same linear ODE: $W' = -pW$

Hessian
$[H_f]_{ij} = \partial^2 f/(\partial x_i \partial x_j)$
$D^2_{\boldsymbol{u}} f = \boldsymbol{u}^\top H \boldsymbol{u}$
eigenvalue signs classify critical points

Exercises

1. Let $\boldsymbol{F}(x, y) = (xy, e^{x-y})$. Find $D\boldsymbol{F}(1, 0)$ and its determinant.
2. Using the Jacobian from Exercise 1, estimate the output change caused by $\boldsymbol{h} = (0.01, -0.02)$.
3. Derive the polar-coordinate Jacobian determinant and use it to recover the area of a disk of radius R.
4. Compute $\mathcal{W}(1, t, t^2)$ and $\mathcal{W}(e^t, e^{-t})$.
5. Find the Hessian of $f(x, y) = x^2 - 4xy + y^2$ and classify the origin.
6. Explain in one sentence why $\det H = 0$ and $W = 0$ require more context than the corresponding nonzero tests.

Answers

1. $D\boldsymbol{F}(1, 0) = \begin{bmatrix} 0 & 1 \\ e & -e \end{bmatrix}$ and $\det D\boldsymbol{F} = -e$; local orientation is reversed and area scales by e.
2. $D\boldsymbol{F}\,\boldsymbol{h} = (-0.02, 0.03e)^\top$.
3. $\det D\boldsymbol{T} = r$ and $\int_0^{2\pi} \int_0^R r \, \mathrm{d}r \, \mathrm{d}\theta = \pi R^2$.
4. $\mathcal{W}(1, t, t^2) = 2$ and $\mathcal{W}(e^t, e^{-t}) = -2$.
5. $H = \begin{bmatrix} 2 & -4 \\ -4 & 2 \end{bmatrix}$ has eigenvalues 6 and -2, so the origin is a saddle.
6. Zero can mean degeneracy or only failure of the present test; higher-order behavior or structural assumptions decide what follows.

Further reading

T. M. Apostol, *Calculus, Volume II*; G. Strang, *Linear Algebra and Learning from Data*; W. E. Boyce, R. C. DiPrima, and D. B. Meade, *Elementary Differential Equations*; J. Nocedal and S. Wright, *Numerical Optimization*.

Differentiate locally. Assemble carefully. Read the geometry.